```
In[∘]:= Quit
```

```
In[1]:= $Assumptions = m > 0 && τ > 0 && k > 0 && T > 0 && m ∈ Reals && τ ∈ Reals
```

Out[1]= $m > 0 \,\&\&\, \tau > 0 \,\&\&\, k > 0 \,\&\&\, T > 0 \,\&\&\, m \in \mathbb{R} \,\&\&\, \tau \in \mathbb{R}$

Distribución de Maxwell para la rapidez

In[2]:= $PM[v\_] := 4\,\pi \left(\frac{m}{2\,\pi\,k\,T}\right)^{3/2} \mathrm{Exp}\left[-\frac{m\,v^2}{2\,k\,T}\right] v^2$

Confirmando la normalización

```
In[3]:= Integrate[PM[v], {v, 0, Infinity}]
```

Out[3]= 1

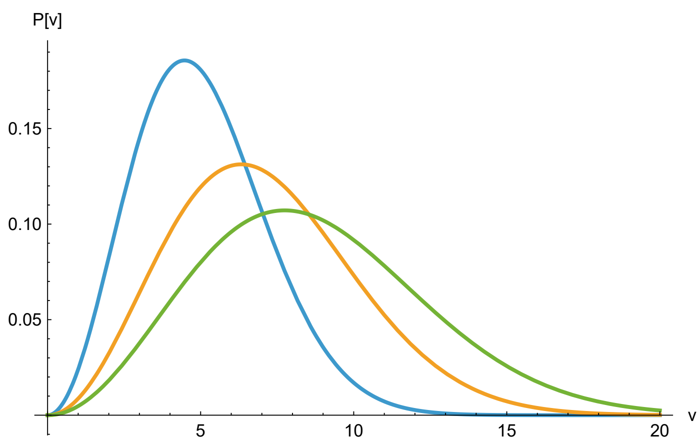
En términos de τ=k T

In[4]:= $PM[v\_, m\_, \tau\_] := 4\,\pi \left(\frac{m}{2\,\pi\,\tau}\right)^{3/2} \mathrm{Exp}\left[-\frac{m\,v^2}{2\,\tau}\right] v^2$

Gráficas para frío τ=10 (azul), intermedio τ=20 (naranja) y más caliente τ=30 (verde)

```
In[5]:= Plot[{PM[v, 1, 10], PM[v, 1, 20], PM[v, 1, 30]}, {v, 0, 20}, AxesLabel → {"v", "P[v]"}]
```

Out[5]=



Rapidez máxima

```
In[6]:= D[PM[v, m, τ], v] // Simplify
```

Out[6]= $-\frac{e^{-\frac{m\,v^2}{2\,\tau}}\, m^{3/2} \sqrt{\frac{2}{\pi}}\, v \left(m\,v^2 - 2\,\tau\right)}{\tau^{5/2}}$

In[7]:= $vmax[m\_, \tau\_] = \mathrm{Sqrt}\left[\frac{2\,\tau}{m}\right]$

Out[7]= $\sqrt{2}\,\sqrt{\frac{\tau}{m}}$

Rapidez promedio <v>

In[8]:=
```
Integrate[v PM[v, m, τ], {v, 0, Infinity}] // Simplify // Refine
```

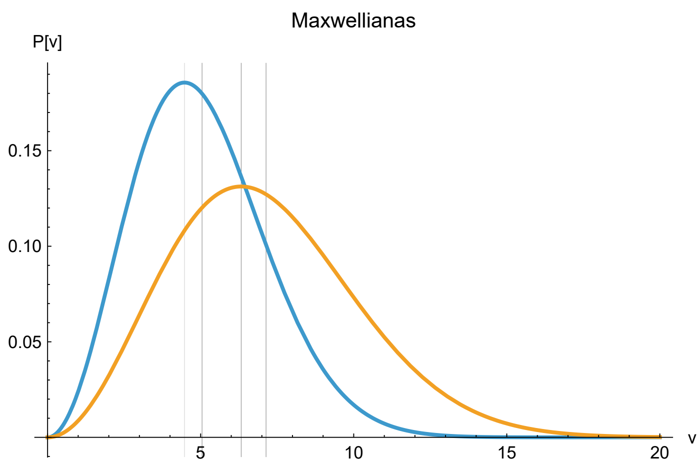
Out[8]= $2\sqrt{\frac{2}{\pi}}\sqrt{\frac{\tau}{m}}$

In[9]:= $\text{vmean}[m\_, \tau\_] = 2\sqrt{\frac{2}{\pi}}\sqrt{\frac{\tau}{m}}$

Out[9]= $2\sqrt{\frac{2}{\pi}}\sqrt{\frac{\tau}{m}}$

In[12]:=
```
Plot[{PM[v, 1, 10], PM[v, 1, 20]}, {v, 0, 20},
 GridLines → {{vmax[1, 10], vmean[1, 10], vmax[1, 20], vmean[1, 20]}},
 PlotLabel → "Maxwellianas", AxesLabel → {"v", "P[v]"}]
```

Out[12]=



Raiz cuadrática media Sqrt[< v^2>]

In[11]:=
```
(Integrate[v^2 PM[v, m, τ], {v, 0, Infinity}])^1/2 // Simplify // Refine
```

Out[11]=

$\sqrt{3}\sqrt{\frac{\tau}{m}}$

Las tres velocidades características vmax, <v>, $\sqrt{(< v^2>)}$ son del mismo orden, pues van como la raiz cuadrada de la temperatura $(T)^{1/2}$